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Automated inventory reorder and sales transparency system for small retailers using deep learning.

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ABSTRACT

Small convenience retailers and independent wholesale distributors operate under tight margins where inventory misallocations severely degrade profitability. Standard manual reordering practices suffer from high inventory records, manual errors and friction losses. This paper presents an end-to-end, two-stage automated replenishment architecture. Stage 1 implements a multi-SKU demand forecasting framework comparing global machine learning models (XGBoost, LSTM) against univariate statistical baselines (SARIMAX). Predictions are combined via a dynamic Inverse WAPE (Weighted Absolute Percentage Error) Ensemble model. Stage 2 integrates automated optical character recognition (OCR) invoice parsing, closed-loop stock accounting, asymmetric top-SKU safety buffering, and discrete sleeve lot-sizing. Empirical results demonstrate that Ensemble pipeline (R^2 = 0.7183, test RMSE = 2.5969) outperform other models by learning cross-SKU patterns, while the Stage 2 replenishment pipeline resolves the automated weekly replenishment gap.

INTRODUCTION

For a mid-sized wholesale distributor generating $10 million in annual revenue, manual operational inefficiencies account for approximately $8,800 per day in absorbed operational friction, translating to an annual financial loss of approximately $2.2 million. Across the United States, where more than 150,000 convenience stores operate—representing over 80% of total industry store counts—implementing automated, data-driven replenishment architectures presents a significant opportunity for margin preservation and operational cost reduction (IHL Group, 2025). DeHoratius and Raman (2008) highlights how over 65% of inventory records disagree with actual inventory counts, representing the failure of human-reliant inventory logging. Empirical research from US retail and wholesale distribution reveals that operational friction costs (manual entry errors, stockouts, overstock, and labor (Table 1. Friction costs for Convenience stores and Wholesale vendors) represent significant profit drains (IHL Group, 2025). This research develops an automated stock replenishment architecture based on SKU forecasting which will resolve the following friction costs faced by convenience stores and vendors (Table 1. Friction costs for Convenience stores and Wholesale vendors):

Convenience store		Vendor	
Friction cost	**Description**	**Friction Cost**	**Description**
Stockout Revenue Losses	Out-of-stock items cost North American food and convenience retailers $15B to $20B (3.0% to 5.9%) annually (IHL Group, 2025).	**Labor/Transport Costs**	Maintaining full-time sales reps (salaries, vehicles, and travel) accounts for a significant volume of friction costs.
Excess Inventory Costs	Overstocking from planning and supplier mismatches incurs holding costs of 1.5%–2.5% of store revenue (IHL Group, 2025).	**Manual Entry Errors**	Manual order entry errors constitute one-third of total frictional supply chain costs.

Table 1. Friction costs for Convenience stores and Wholesale vendors

Deploying automated architectures and ML streamlines enable retailers to achieve up to 2.5x higher profit growth than competitors (IHL Group, 2025). Replacing the above manual work with single automated multi-SKU forecasting and replenishment architecture resolves all the operational leaks for both retailers and their vendors by providing them a competitive edge in the industry.

FOUR-STEP WORKFLOW

According to Sloan, Gillette, and Padhi (2023), managing demand forecasting across multi-tier product hierarchies requires bridging rigorous statistical architectures with operational flexibility. This research implements a hybrid pipeline that maps following four-step workflow (Figure 1. Four-step Workflow):

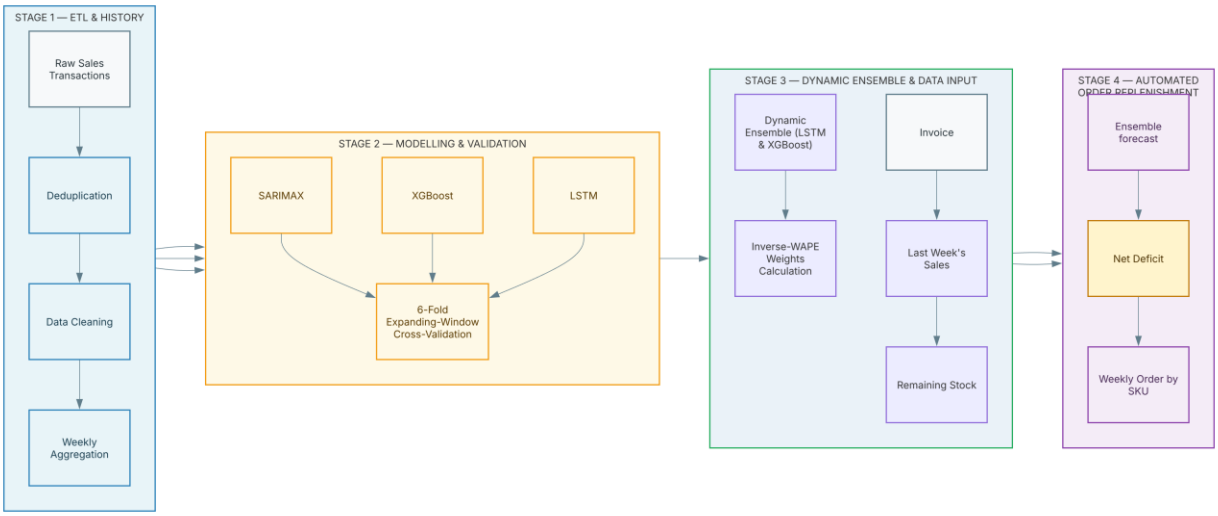


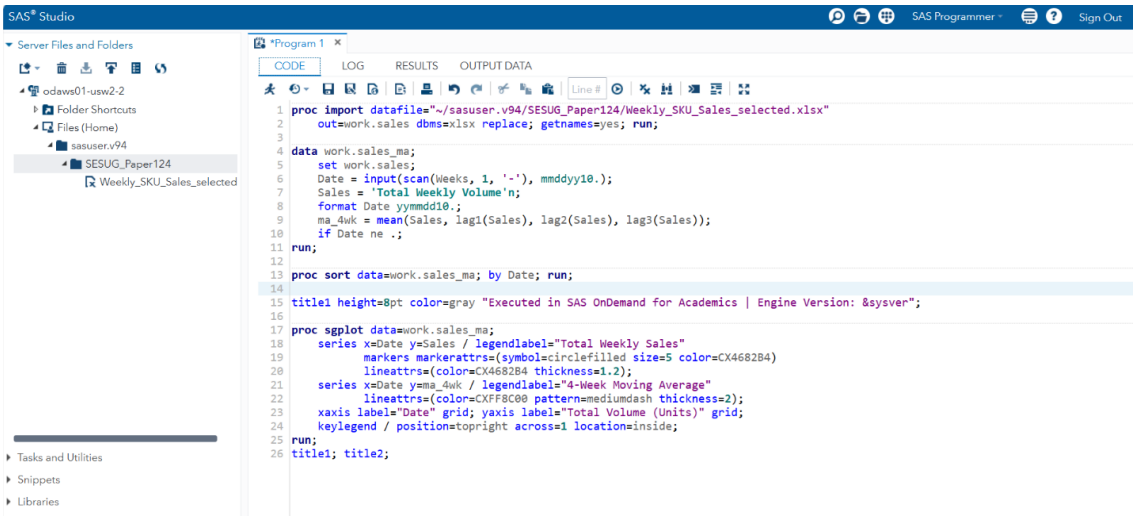
Figure 1. Four-step Workflow

FORECASTING ARCHITECTURE

DATA SOURCE AND ANALYSIS

The dataset for this study is derived from individual sales transaction records captured at a convenience store located in Kansas City, Missouri, spanning from August 15, 2023, to August 16, 2026. This raw dataset includes columns like transaction IDs, timestamps, product names, departments, quantity sold in units, and sales prices. While the data provides granular visibility, direct application of high-frequency timestamped data to weekly inventory replenishment models carries excessive noise with intermittent demand spikes (Chen et al. 2000).

To align the data with standard weekly replenishment cycles, the transactional records were aggregated into weekly total sales volumes for each Stock Keeping Unit (SKU). Aggregating raw individual transactions into weekly intervals establishes a stable historical baseline essential for accurate demand forecasting (DeHoratius and Raman 2008).



Display 1. SAS OnDemand for Academics (Engine Version 9.4)

As shown in the implementation script above (Display 1. SAS OnDemand for Academics (Engine Version 9.4) the time-series pipeline was executed in SAS OnDemand for Academics (Engine Version 9.4). The output visualization (Figure 2. Weekly SKU sales with 4-week moving average (08/15/23 - 08/16/26)) was generated using the PROC SGPLOT procedure.

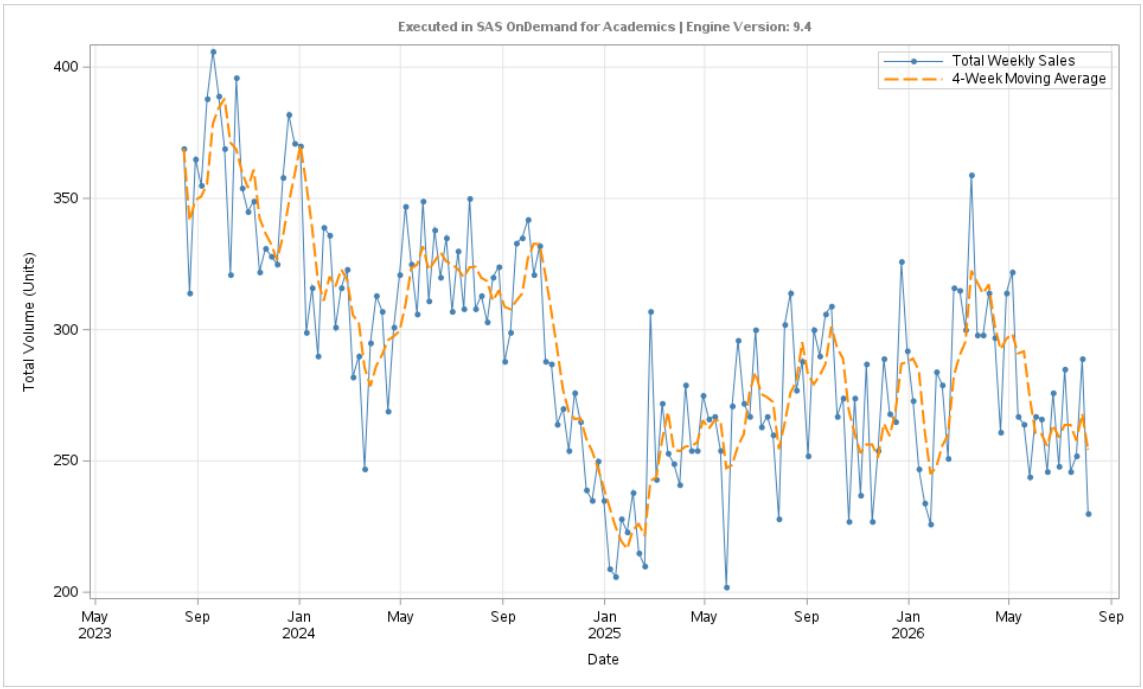


Figure 2. Weekly SKU sales with 4-week moving average (08/15/23 - 08/16/26)

Starting from December 2024, weekly sales experienced a sharp, permanent downward shift with the 4-week moving average dropping from peaks of 300–350 units down to 200–250 units due to a new nearby store opening (Figure 2. Weekly SKU sales with 4-week moving average (08/15/23 - 08/16/26)).

FORECASTING MODELS

- Seasonal autoregressive integrated moving average with exogenous variable (SARIMAX) - Because p-value = 0.073 (which is > 0.05) for Augmented Dickey-Fuller test our data is not stationary and requires differencing (d=1). Python's Auto_arima function with differencing fixed at

d=1, was used to make an automated search across parameter combinations and selecting the optimal model based on the lowest Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). The automated search identified the best-performing SARIMAX specification as order (0,1,1), $(0,1,1)_{52}$, yielding an AIC of 386.76 and a BIC of 391.83.

- Extreme Gradient Boosted Decision Trees (XGBoost) – XGBoost, has exceptional computational efficiency with built-in regularization, and feature interactions across diverse data points (Chen, 2016). To optimize predictive performance of XGBoost Optuna (Automated framework for hyperparameter tuning) was executed on each of 6-fold cross-validation splits to optimize key hyperparameters by minimizing squared errors across weekly SKU observations (Table 2. XGBoost Hyperparameter values).

Hyperparameter	Range	Selection Rationale
n_estimators	[50, 100, 150, 200, 250, 300, 350, 400]	More trees allow us to learn complex patterns, but too many risks adding training noise
learning_rate	0.01 to 0.1	How much weight each new tree adds to the overall forecast. Gradual (0.01) vs aggressive (0.1) learning.
max_depth	2 to 6	How tall each tree can grow. Too shallow tree risks underfitting vs too deep tree risks overfitting
subsample	0.6 to 1.0	When building each tree, the model uses 60% -100% of historical rows and is not overreliant on specific sales periods
colsample_bytree	0.6 to 1.0	When building each tree, the model uses 60% -100% of feature columns and not overreliant on one feature

Table 2. XGBoost Hyperparameter values

- Long Short-Term Memory (LSTM) – LSTM is recurrent neural network architecture designed to capture sequential patterns and long-term temporal dependencies in time-series forecasting. To optimize performance, Optuna was used across 6 – fold cross-validation splits (Table 3. LSTM Hyperparameter values).

Hyperparameter	Range	Selection Rationale
hidden_dim	{16, 32, 64}	Too small memory size prevents learning complex relationships (underfitting) but larger memory size risks overfitting
lr (Learning Rate)	0.0001 to 0.01	Too high lr risks unstable training, too low makes convergence very slow
num_layers	1 to 2	Limiting number of recurrent layers to 1-2 prevents the model from overfitting in short dataset
dropout	0.1 to 0.4	Randomly deactivates 10%-40% of recurrent connections during training, forcing network to learn robust paths
seq_length	{3, 4}	Corresponds to 3 to 4 weeks of historical lookback window (roughly one month) to capture monthly demand seasonality trends.

Table 3. LSTM Hyperparameter values

6-FOLD EXPANDING WINDOW CROSS-VALIDATION

To prevent data leakage and temporal bias, a 6-fold expanding window cross-validation framework was implemented where training data strictly precedes validation data across all splits (Table 4. Performance comparison of Models across 6-Fold expanding window cross-validation). Performance deterioration in Fold 5 (April 21, 2026 – June 09, 2026) occurred because all models struggled to generalize to an unusually volatile rise in quantities sold, causing XGBoost and LSTM models to take a

temporary accuracy hit. In Fold 6, performance was fully restored as underlying sales velocity matched the prevailing regime.

Fold	Training Period	Validation Period	SARIMAX (R^2)	XGBoost (R^2)	LSTM (R^2)
Fold 1	08/15/23 - 09/02/25	09/09/25 - 10/28/25	0.6263	0.7492	0.7385
Fold 2	08/15/23 - 10/28/25	11/04/25 - 12/23/25	0.6378	0.7258	0.7203
Fold 3	08/15/23 - 12/23/25	12/30/25 - 02/17/26	0.6000	0.6864	0.7171
Fold 4	08/15/23 - 02/17/26	02/24/26 - 04/14/26	0.6139	0.7112	0.6666
Fold 5	08/15/23 - 04/14/26	04/21/26 - 06/09/26	0.6000	0.5992	0.6500
Fold 6	08/15/23 - 06/09/26	06/16/26 - 08/04/26	0.6500	0.7500	0.7500

Table 4. Performance comparison of Models across 6-Fold expanding window cross-validation

INVERSE-WAPE DYNAMIC ENSEMBLE (LSTM AND XGBOOST)

To produce optimal predictions, outputs from both LSTM and XGBoost are combined using a dynamic Inverse-WAPE weighting mechanism, expanding upon the foundational forecast combination principles established by Bates and Granger (1969) and conditional weighting strategies outlined by Aiolfi and Timmermann (2006). Models with lower WAPE are assigned proportionally higher combination weights, mitigating individual model volatility and stabilizing retail forecasts.

$$\text{XGBoost}_{\text{Weight}} = \frac{(LSTM\ WAPE)}{(XGBoost\ WAPE + LSTM\ WAPE)} = \frac{34.41}{(37.77+34.41)} = 0.4767$$

$$\text{LSTM}_{\text{Weight}} = \frac{(XGBoost\ WAPE)}{(XGBoost\ WAPE + LSTM\ WAPE)} = \frac{37.77}{(37.77+34.41)} = 0.5233$$

The model with the lower WAPE receives the higher weight in Ensemble model. For example, for KOOL BLACK BOX 100s XGBoost $_{\text{WAPE}}$ (37.77%) is higher than LSTM $_{\text{WAPE}}$ (34.41%). Consequently, LSTM is assigned a higher weight (LSTM $_{\text{weight}}$ = 0.5233) than XGBoost (XGBoost $_{\text{weight}}$ = 0.4767) **Table 5. Inverse-WAPE weighting and Ensemble prediction (08/03/26 - 08/10/26)**

SKU (Stock Keeping Unit)	Actual Volume (Units)	XGBoost WAPE (%)	LSTM WAPE (%)	XGBoost Weight Ensemble	LSTM Weight Ensemble	XGBoost Prediction (Units)	LSTM Prediction (Units)	Ensemble Prediction (XGBoost/ LSTM)
KOOL BLACK BOX 100S	21	37.77	34.41	0.4767	0.5233	12.45	13.08	12.78
MARL BOX KING	16	17.11	13.44	0.4399	0.5601	15.16	15.63	15.42
MARL NXT BOX KING	16	6.97	18.53	0.7267	0.2733	15.1	12.92	14.50
24/7 RED 100 BOX	12	68.99	58.36	0.4583	0.5417	12.06	8.95	10.38
MARL GOLD BOX KING	11	25.57	25.42	0.4985	0.5015	11.51	13.1	12.31
NEWPORT SOFT 100'S	11	27.17	25	0.4791	0.5209	15.42	15.26	15.34
DECADE MENTHOL 100	9	36.14	40.89	0.5309	0.4691	14.69	15.11	14.89
MARL MEN BOX KING	9	36.93	34.52	0.4831	0.5169	6.42	5.86	6.13
KOOL BLACK BOX KINGS	8	32.12	37.07	0.5358	0.4642	10.72	5.83	8.45
NEWPORT SOFT KING	8	30.85	64.72	0.6772	0.3228	8.37	14.92	10.48

Table 5. Inverse-WAPE weighting and Ensemble prediction (08/03/26 - 08/10/26)

According to Hyndman and Koehler (2006) WAPE provides a balanced, scale-free foundation. Unlike MAPE (Mean Absolute Percentage Error) which breaks down with zero-demand intermittent SKUs due to division by zero or RMSE (Root Mean Squared Error) which heavily penalizes high-magnitude series while ignoring low-volume items, the Weighted Absolute Percentage Error (WAPE) provides a balanced, scale-free foundation.

MODEL COMPARISON

Although SARIMAX demonstrates “good generalization” with test R^2 = 0.6147 explaining 61.47 % of the variance in weekly sales volume and test RMSE = 3.4615 indicating that on average, SARIMAX's weekly sales predictions deviated from the actual SKU sales volume by roughly 3.46 units but it’s still relatively inferior performance to machine learning models (Table 6. Model comparison). Among the standalone models, XGBoost proved to be the most accurate, achieving strong generalization with a test R^2 = 0.7057 and a test RMSE = 2.6544.

Model	RMSE		R^2		MAPE (%)		WAPE (%)		Diagnosis
	Train	Test	Train	Test	Train	Test	Train	Test	
Ensemble (XGB & LSTM)	2.5295	2.5969	0.8180	0.7183	48.01	60.11	38.73	47.50	**Strong Generalization**
XGBoost	2.6501	2.6544	0.8002	0.7057	52.69	61.23	41.94	47.70	**Strong Generalization**
LSTM	2.5285	2.7101	0.8181	0.6932	46.40	63.51	37.08	49.76	**Good Generalization**
SARIMAX	3.1621	3.4615	0.6840	0.6147	56.29	65.62	45.26	51.89	**Good Generalization**

Table 6. Model comparison

The Dynamic Inverse-WAPE Ensemble (XGB+LSTM) achieved the strongest generalization, with test R^2 = 0.7183, test RMSE = 2.5295 and test WAPE = 47.50 (Table 6. Model comparison). By dynamically weighing LSTM and XGBoost based on each model’s inverse WAPE, the ensemble architecture successfully dampened model-specific volatility, minimized out-of-sample error, and protected against overfitting on volatile retail SKU data.

SCATTER PLOT EVALUATION

Evaluation of Actual versus Predicted scatter plots across every model shows a dense cluster above the X=Y red dashed line indicating that models frequently overpredicted low-volume SKUs for up to 5 units, struggling with slow-moving items (Figure 3. Actual vs Predicted X=Y (08/03/26 - 08/10/26)). On the other hand, models predominantly underpredicted high-volume SKUs falling below the actual X=Y red dashed line

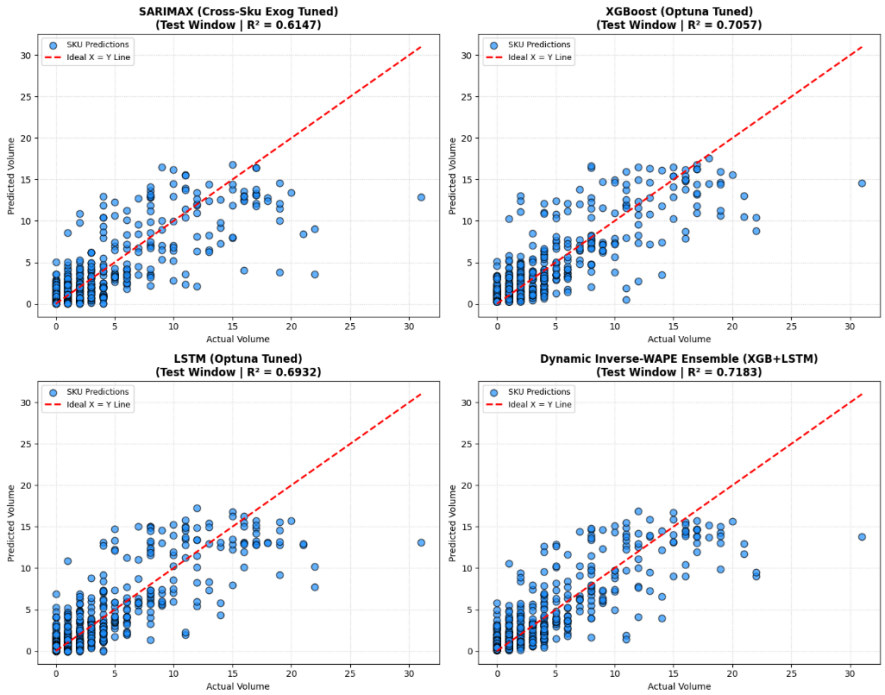


Figure 3. Actual vs Predicted X=Y (08/03/26 - 08/10/26)

IMPLEMENTATION: AUTOMATED OPERATIONAL REPLENISHMENT ENGINE

DYNAMIC CROSS VALIDATION AND ENSEMBLE WEIGHT OPTIMIZATION

To maintain predictive accuracy amidst structural shifts in retail demand, the pipeline accumulates two months of actual sales data before establishing a new validation window (Fold 7). During this interval, the system re-tunes the core hyperparameters for both the XGBoost and LSTM models to ensure optimal structural alignment with the prevailing sales regime. Utilizing these recalibrated hyperparameters, the pipeline computes the updated Weighted Absolute Percentage Error (WAPE) for each model across all SKUs, deriving new inversely proportional weights to generate the final ensemble prediction, consistent with the methodology established in the Inverse-WAPE Dynamic Ensemble section.

Furthermore, this architecture maintains a fully automated, continuous operational loop. Every subsequent week, the Stage 2 pipeline automatically ingests the past week's actual sales data into the historical training dataset to construct refreshed expanding cross-validation folds (Fold 8 and onward). This continuous ingestion dynamically updates the ensemble weights and recalibrates point forecasts to reflect evolving consumer behavior without requiring manual model retraining (Bates and Granger, 1969).

AUTOMATED ORDER GENERATION

The pipeline uses the generated ensemble predictions to compute net deficits for each SKU and construct final purchase orders for the next week (Table 7. Automated weekly order generation). The system ingests foundational inventory stock counts (which are entered exclusively during initial system setup and maintained dynamically thereafter), current week actual sales data, and newly received delivery invoices processed via EasyOCR to automate reconciliation and eliminate manual entry latency (Raman, DeHoratius, and Ton, 2001)

SKU	Beginning Stock (08/03/26)	New Delivery (Invoice)	Past 7-day Sales	Ending Stock (08/10/26)	Ensemble Prediction (08/10/26 - 08/16/26)	Buffer	Net Deficit	New Order for Next Week (Sleeves)
KOOL BLACK BOX 100S	15	10	18	7	20	5	18	2
DECADE MENTHOL 100	16	0	16	0	14	5	19	2
MARL BOX KING	4	10	14	0	12	5	17	2
NEWPORT BOX KING	3	10	13	0	14	5	19	2
NEWPORT SOFT 100'S	4	10	13	1	10	5	14	1
NEWPORT SOFT KING	4	10	12	2	9		12	1
MARL NXT BOX KING	18	0	12	6	10		9	1
DECADE GOLD 100	11	0	11	0	9		14	1
24/7 RED 100 BOX	11	0	11	0	13		18	2
MARL GOLD BOX KING	11	0	11	0	15		20	2
ECHO RED KING BOX	2	10	9	3	11		13	1

Table 7. Automated weekly order generation

As illustrated in Table 7. Automated weekly order generation, the pipeline organizes SKU data across each column starting with beginning stock quantity, new deliveries (according to invoices), past 7-day sales, ending stock quantity, ensemble predictions, (flexible) SKU buffer quantities, net deficits, and final reorder quantities. The pipeline addresses safety buffering specifically to high-velocity top-tier SKUs, to protect revenue-critical items against sudden demand surges (IHL Group, 2025). Finally, at the end of each weekly cycle, the code automatically computes and saves the remaining ending inventory stock to serve as the beginning stock baseline for the subsequent week, ensuring that future weekly runs require only the incoming invoice and sales logs as inputs (IHL Group, 2025).

CONCLUSION

By addressing supply chain distortions highlighted by IHL Group (2025), this research successfully integrates advanced predictive analytics with real-life physical retail operations. Diagnostic evaluation across SKUs confirms that ensembling predictions from XGBoost and LSTM using the dynamic Inverse-WAPE technique yields the strongest generalization by achieving the highest level of R^2 (0.7183) while reaching the lowest test RMSE (2.5969) and lowest WAPE (47.50%). Dynamic weighting of LSTM and XGBoost based on inverse WAPE across every SKU enables the system to dampen model-specific volatility and guards against overfitting by efficiently capturing complex non-linear patterns across diverse product categories. Crucially, as new data becomes available across expanding cross-validation folds, hyperparameters are dynamically re-tuned and model combination weights are recalculated, ensuring the ensemble continuously adapts to shifting demand patterns over time.

Ultimately, bridging the validated Ensemble model with the invoice and inventory ledgers produces next week's order by eliminating manual reconciliation bottlenecks and friction costs discussed in IHL Group (2025). This seamless synthesis of predictive modeling and physical inventory tracking empowers mid-sized wholesalers and small convenience retailers who cannot afford expensive automated replenishment software to eliminate millions in annual friction costs, preserve operating margins and achieve sustainable competitive advantage.

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